

NAMIE Nomvula Charlotte Mkhweni

COURSE DATA ANALYTICS AND DATA SCIENCE

SECTION A 'Data Fundamentals

1 Database is a digital place where a collection of data is stored in an organised way so that it can be accessed easily. Databases are essential in modern world as they eliminate the use of physical file storage and enables users to retrieve huge amounts of data in a simplified way.

Examples : Banks
: Schools.

2. Datawarehouse is a storage system that is concentrated in one place with data from many different places, within an organisation. Datawarehouse is for analysing data.

Differences between database and datawarehouse

Datawarehouse	Database
- It analyses data - It uses OLAP (Online analytical Processing) - Deals with simple complex queries - It is at head office.	- It is for reporting data - It uses OLTP (Online Transactional Processing) - Deals with simple transactional queries - It is at Branch level

a company would use datawarehouse to maintain historic data records and it keeps data very secured.

3. Data Lake stores all kinds of data, i.e. unstructured, raw and unprocessed, whereas data warehouse stores structured and processed data. Business would prefer to make use of a data lake when there is massive amounts of data that is not structured, processed and is raw that needs to be stored.

4. Data mart is a data storage place within the data warehouse, it is specialised and focused on a certain team and or purpose. Data warehouse has all the organisation's data about different topics ~~whereas~~ whereas data mart is focused on a specific and specialised data within an organisation's data warehouse.

Data mart can be used by marketing team, sales staff, finance staff and business analysts.

5. Structured data - the type of information that is stored in an organised manner and can be easily retrieved. This data is stored in rows and columns. e.g. Spreadsheets.

Unstructured data - The type of data that does not have a format for storage. It is not organised. e.g. text messages.

Semi-structured data - The type of data that is stored using graphs, tags and markers. e.g. web pages.

6. Data modeling is a detailed plan of data before it can be put in a database. Data modeling prevents errors and ensures accurate data is captured.

Data models

Conceptual - It says what kind of data does a business need.

Logical - It builds on the ~~need~~ data needs of the business through using a structure.

7. Data mining is a way of finding patterns that are hidden and other important information from large data sets for decision making using machine learning. It makes use of algorithms to ~~get~~ information from both structured and unstructured data.

examples : Retail shops - customer buying history to study which products are constantly purchased.
Banks - to monitor transaction history of clients.

8. Database Management System is a software tool that makes it easy to organise data into the database.

MySQL - it is used in web development.

Suitable Use Case -> web applications & E-commerce

PostgreSQL - Open-source relational database system, it is free to use

Suitable Use case -> AI applications

MongoDB - Database that stores data in JSON-like documents

Suitable Use Case : Content Management

Oracle - Mainly used for core banking systems

Suitable Use case -> Government Systems

9. ETL (Extract, Transform, Load) - It is a process where data is moved from where it is created to where it is analysed.
Extract - Data that is raw is extracted from different sources.
Transform - Extracted data is then processed, cleaned and modified.
Load - Then cleaned, processed and modified data is now put into a data system.

The importance of ETL in data analytics is that it ensures that data is structured, consistent and cleaned.

10. JSON - Used for the storage of structured data.

CSV - used to store tabular data in spreadsheets or databases.

Parquet - Organises data using columns

Avro - It stores data using row-based format.

~~SECTION B: Artificial Intelligence & Machine Learning.~~

References

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 - ④ <https://www.ibm.com>
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SECTION B Artificial intelligence and Machine learning

11. Artificial Intelligence - It is the kind of technology that imitates human intelligences through computers and machines.

Narrow AI > It is trained to do one specific task and cannot transfer knowledge to unrelated tasks.
e.g. META

General AI > It can learn, reason and do cognitive tasks over multiple domains. It basically mimics human-level intelligence.
e.g. WALL-E.

12. Machine learning is a way in which computers are taught to learn from examples rather than giving them step-by-step instructions on how to respond to prompts.

With traditional programming, you are the one writing the rules and with machine learning the computer learns the rules from data.

Supervised - This type of machine learning has been given algorithm data labeled with correct answers.

Unsupervised - It has to find patterns and structures on its own with no answers.

Reinforcement - It learns to maximise rewards through trial and error method.

12. Deep learning is learning that uses artificial neural networks that has many layers to learn from directly from raw data. Deep learning makes use of neural networks to do machine learning and through machine learning AI systems are built.

Neural networks - They are computing systems that work in a similar way with neurons in human brain. They work by processing data by layers, then a prediction comes out then they learn by adjusting the data that has put in raw.

14. Natural Language Processing is a part of AI that assists computers to understand, interpret, and generate human language the same way humans speak it and write it.

e.g (i) Chatbots

(ii) Emails.

(iii) Search engines.

15. Large Language Model - It is a computer that is trained through AI on huge amounts of texts to predict and generate human-like language. Standard Machine Learning model is narrow and specific. LLM is broad and flexible with language.

Examples (i) GPT - For coding

(ii) Gemini - For generating texts, images, videos and audio.

16. Generative AI is the type of AI that has the capabilities of creating new content instead of analysing existing data. It can learn patterns, generate and create new content.

(i) Chat GPT - for writing emails, brainstorming, coding and answering questions.

(ii) Claude - Writing and editing long documents, coding and research help.

(iii) Veo - Generating short videos from texts or editing existing videos.

17. Training dataset is a set of labelled and unlabeled data that is used to teach a machine to recognise patterns and be able to make predictions. Quality of data training is important to ensure accuracy. If training data is biased or incorrect the AI can deliver unfair results and make wrong predictions.

18. Computer Vision is a type of AI that enables the machine to analyse and understand images and videos the same way as humans do. Machines see through a camera capturing pixels as numbers, process them and then interpret the data.

Uses : Healthcare > CT scans

Security > Surveillance cameras

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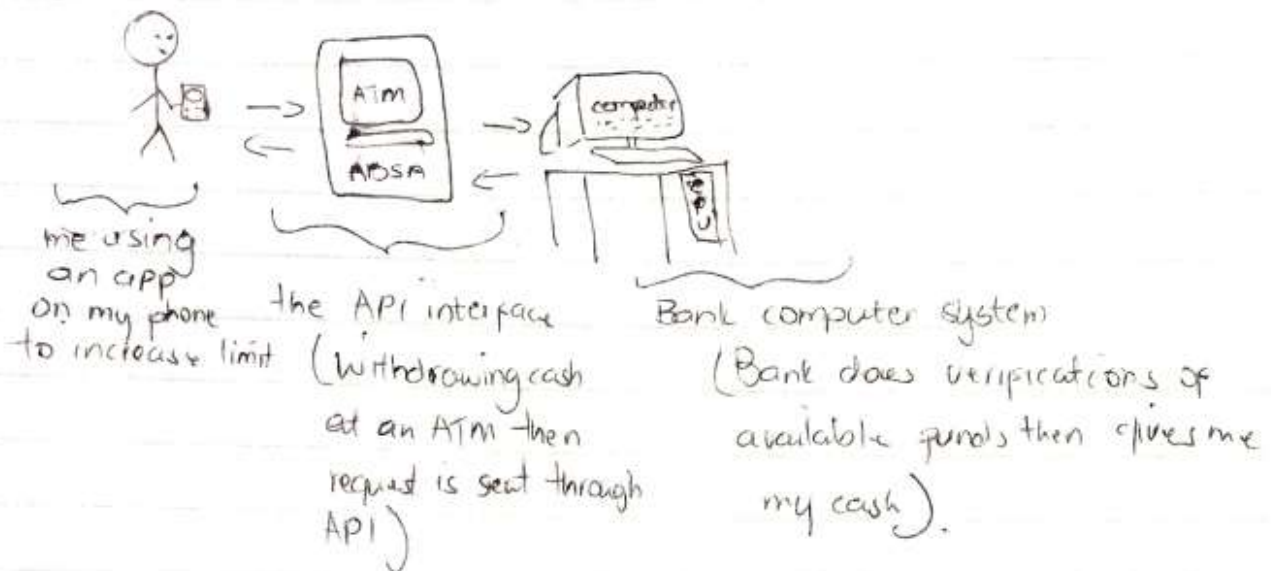
<https://www.ibm.com>

Google developers

<https://cloudfactory.com>

SECTION C: APIs, TOOLS and Developer Concepts

19. Application Programming Interface is a set of rules that allows software programs to communicate with each other.



APIs are important in modern software as they save time connects systems then data is shared in real time.

20. API key is a secret code that is responsible for authentication and authorization on an app in order to use API. It must be stored by secret managers and not in public repos. If it becomes exposed to public, attackers can misuse it and access data, that is sensitive.

21. REST API is an Application Programming Interface that follows the REST architectural style to let different systems to communicate using standard web methods over HTTP.

REST - stands for REpresentational State Transfere.

HTTP methods : Post > creates new resources

PUT > Updates/replaces an existing resource completely.

DELETE → Removes resources at a specific URL

22. JSON (Java Script Object Notation) is a text base-data format for storing and exchanging data using key-value pairs and arrays. It is used when working with APIs and data exchange because it is simple, fast and works every where.

```
{ "product": "Notebook", "price": 10.00, "in stock": true }
```

23. Prompt engineering is learning a way in which you can ask AI questions the correct way so that it gives you the kind of answer that you really want. It is important because a good prompt gives out accurate results faster.

Example 1 : Write about Aloe vera (poorly written prompt).
: Write a brief summary about the health benefits of aloe vera (improved version)

Example 2 : I need the colour red (poorly written prompt)
: What colours do I need to mix to get the colour red, please provide accurate measurements in a wall paint, (Improved version).

24. MAP function is a function that takes a list of numbers, objects, strings etc and applies the same operation to every item in it then later returning a new list with the results. MAP function saves users from writing a manual for a loop everytime a need for data transformation arises.

JavaScript code

```
const nums = [5, 10, 15];  
const new-nums = nums.map(x => x + 1);  
console.log(new-nums); // [6, 11, 16]
```

25 Cloud computing makes use of servers, storage and software over the internet instead of my computer.

IaaS - Gives out raw computing Infrastructure/tools
e.g Amazon EC2.

PaaS - Where an app is uploaded and the platform handles servers and set up. e.g Google App Engine.

SaaS - A ready-made online software, no installation needed.
e.g google docs.

26 Version control System is a system that tracks and saves changes to files over time. It is widely used cause of it is efficient, works offline. Git is the most used version control tool.

Commit - A saved snapshot of your project with a short note of changes

Branch - Another separate copy of a project to test changes in a safe way.

27. Jupyter Notebook is an interactive web app for writing codes mixed with texts and charts. It is mostly used with Python. It lets data analysts and Data Scientists to run codes step-by-step, see results in an instant and it also combines codes, data and explanations in one place

28. Python is a programming language used for web applications, data analysis, automation, AI and more. It is a popular programming language because of its simple syntax and a massive ecosystem of libraries for maths, data and ML. It is widely used in ML community.

1. Python Libraries:
- (i) Pandas > Handles tables, and time series data.
 - (ii) Matplotlib > Creates static plots and charts.
 - (iii) Numpy > Provides fast arrays and math functions.
 - (iv) Scikit learn > machine learning library with tools for classification, regression and model evaluation.
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SECTION D : Data Ethics, Governance & careers.

29. Data privacy is the right for individuals that allows them to control on how their personal data is shared, used and collected. It is essential as it protects users personal information from being misused, stolen or shared without consent.

GDPR is General Data Protection Regulation. It is the main data protection law of the EU. It applies to EU member states, EEA countries.

30. Data Ethics is a set of principles that stipulate how data should be collected, used and shared in a way that respects people's

examples of unethical use of data

example 1: A loan algorithm that was trained on old data where certain groups based off their race get rejected.

Why is it problematic: Unfair and discriminatory

Example 2: A health app collecting sensitive medical information and sells it to advertisers, that can identify users

Why is it problematic: Breach of privacy

Example 3: An app reading private chats and using that to advertise things mentioned in your private chats.

Why is it wrong: Privacy is violated.

31. Data governance is a set of rules and roles that decides how an organisation collects, uses and protects its data.

Key components of good data governance Framework

- (i) Data ownership & accountability - responsible for quality, access and compliance.
- (ii) Policies & standards - Rules for data quality, privacy, security retention and naming
- (iii) Data quality management - for checking accuracy, completeness and consistency.
- (iv) Data catalog & metadata - a directory describing what data exists and who owns it.
- (v) Security and compliance - access controls and monitoring to meet laws like GDPR.

32 AI bias is when an AI system gives unfair results towards certain

people because of biased training design. This happens when a model treats past patterns as truth so it repeats and amplifies human prejudices.
 example: Google photos in 2015 labeled black people as gorillas because the training data had too few images of black people.

Consequences = mislabeling caused public backlash and forced Google to fix the system.

33 Role	Main role	Key Skills	Common tools.
Data Analyst	Analyse data	- Statistics, SQL data visualisation	- Excell, SQL, Python Power BI, Tableau.
Data Scientist	Builds models and find patterns	- Statistics, ML Python/R, data cleaning	Python, R, Scikit-learn Jupyter, SQL
Data engineer	Build & maintain data pipelines	- ETL, database, Cloud, data architecture	SQL, Spark, Airflow, AWS/GCP, Azure, Kafka
AI engineer	Deploy AI/ML models into products & systems	Deep learning, MLOps Python, model optimisations	PyTorch, Tensorflow, Docker Hugging face

34 Business Intelligence is using data and Software to analyse past performance to help businesses make informed decisions. They turn raw data into visuals and spot trends quickly.

References

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- (iii) blog.box.com
- (iv) www.salesforce.com
- (v) ~~Tech~~ www.techtarget.com
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SECTION E : Personal Reflection

35. Artificial Intelligence and Machine learning. I got to learn that computers can actually learn on their own than following a set of rules and with more data they can improve.

36. APIs, Tools & Developer concepts was the most difficult to use I relied on google then it redirected my to relevant pages and also used IBM and for best understanding I used chatGPT

37. Data and AI will simplify life, make jobs easier and stress free as it will automate tasks and will handle boring and repetitive stuff.

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